

- 1) Suppose you evaluate the test set after every training epoch, and keep adjusting your model until the test accuracy is the highest.

You run the risk of overfitting, since the test data repeatedly goes into the model optimization process. It's important that the evaluation be done on new data. For instance if you evaluate your model repeatedly against a particular benchmark you'll negate the effectiveness of the benchmark, like the student memorizing the answers from previous tests.

- 2) Why might validation loss begin increasing even though training loss continues to decrease?

Once we start over-fitting the data, we lose generalizability.

- 3) Suppose validation loss is lowest at epoch 18, Training finishes at epoch 40. Which model should you deploy?

Epoch 18. Once validation loss starts increasing, it's a sign that we are over-fitting. Going 22 more times is likely a huge waste of time, compute, and money.

4) Why can stopping earlier produce a model that performs better than a model trained longer.

Models can eventually memorize anything, and they lose their ability to generalize, since once it takes a lot less compute to effectively use a look-up table than to model the actual function.

5) Why does patience in validation loss produce better results than stopping at the first sign?

These are stochastic processes. We need more data to separate noise in our evaluations from actual model performance degradation.

6) Why do you think machine learning separates optimization of parameters and hyper parameters instead of trying to learn everything automatically?

We are trying to explore the solution space of what kind of model produces the best outcomes, and instantiating that by training our parameters.

Eventually both are likely to become automatable, but currently training a particular model is an expensive and time-consuming process.

Changing more than a few parts of a complex system at once makes it difficult to determine what change created what result.